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IMPLEMENTING THE SHANTI ACT

To achieve India's aim of 100 GW of installed nuclear power capacity by 2047, operationalising a robust regulatory architecture for the SHANTI Act is critical

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► **THE SUSTAINABLE HARNESSING** and Advancement of Nuclear Energy for Transforming India (SHANTI) Act marks a pivotal shift in India's approach to nuclear energy by balancing investment needs, energy security, climate commitments and safety imperatives. A robust institutional architecture must be established for the Act to come into effect efficaciously.

Core elements of the operational nuclear ecosystem that must be addressed include an evolution trajectory laid out in the national nuclear policy; incubating applied research; dimensions of safety and technical regulation, licensing, tariff mechanism, financial security and liability, and insurance architecture; institutional capacity, workforce upskilling, and broader enablers such as mining, intellectual property, confidentiality and claims processes.

The Act provides an overarching safety and technical framework, but detailed subordinate legislation is needed to define “prescribed equipment” and “prescribed substances”, identify facilities and activities requiring safety authorisation, and set stage-wise standards across the lifecycle of nuclear operations—

from design and siting to decommissioning. The rules must also prescribe qualifications and training for personnel, identify radioactive substances requiring licensing and a national registry, establish protocols for storage, transport and emergency

In a price-sensitive market like India, implementing such a programme will require significant financial structuring. The central questions are how such capital expenditure can be made viable and creditworthy; how the costs of safety measures for



A PREDICTABLE AND TRANSPARENT LICENSING REGIME MUST DEFINE THE RULES FOR ENTRY INTO, PARTICIPATION IN AND EXIT FROM THE MARKET

disposal. This will align India with global nuclear governance norms.

A predictable and transparent licensing regime must define the rules for entry into, participation in and exit from the Indian market. It should specify conditions for grant of licences, application requirements, fees, performance standards, and triggers for suspension or cancellation, together with clear mechanisms for handling facilities, materials and equipment in such cases.

storage, handling and O&M of the nuclear island—including prescribed equipment and material—will be absorbed; and how nuclear energy can be assured must-run and must-dispatch status in a merit-based grid.

To unlock financing, especially from international investors and insurers, the layered financial protection framework envisaged by the Act must be clearly articulated. This includes prescribing how financial security for non-nuclear damage is